A QUASI-COMPACT MONOMORPHISM OF ALGEBRAIC SPACES WITH INJECTIVE STRUCTURE SHEAF PULLBACKS IS AN ISOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-more-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance, as required for this source: the first line of the pinned chapter file is \input{preamble}, and the native preamble was not staged with the source, so no preamble text lives inside the chapter. The retained preamble used here is the project's single Stacks-wide preamble (the same 217-line / 8116-byte file that the already published chow and etale-cohomology cards use), copied mechanically line for line from sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Not one character was changed, no package was added, no macro was defined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The same-chapter pointers are located at their \label lines inside the pinned chapter file; the cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Existence of an affine etale cover of a quasi-compact algebraic space. Properties of Spaces, Lemma labelled spaces-properties-lemma-quasi-compact-affine-cover; invoked at lines 331-333. Supplies U = Spec(B) with a surjective etale morphism U -> X.
- Quasi-coherent modules on X are descent data for A -> B. Properties of Spaces, Proposition labelled spaces-properties-proposition-quasi-coherent together with Descent, Definition labelled descent-definition-descent-datum-modules; invoked at lines 335-340. Identifies the category of quasi-coherent O_X-modules with DD_{B/A}.
- Descent for modules along a universally injective ring map. Descent, Theorem labelled descent-theorem-descent; invoked at line 349. Reduces DD_{B/A} to the category of A-modules once A -> B is seen to be universally injective.
- fpqc descent of the property of being an isomorphism. Descent on Spaces, Lemma labelled spaces-descent-lemma-descending-property-isomorphism; invoked at lines 358-359. Turns 'becomes an isomorphism after base change by X -> Spec(A)' into 'is an isomorphism'.

Source reference. Original lemma 023G: sources/stacks-chow/tags.txt, source lines 2741–2741. Navigation pointer for \ref{descent-definition-descent-datum-modules} used inside the counted span (line 339); the argument that consumes it is retained in the span.

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Source reference. Original lemma 08WW: sources/stacks-chow/tags.txt, source lines 11284–11284. Navigation pointer for \ref{descent-subsection-descent-modules-morphisms} used inside the counted span (line 340); the argument that consumes it is retained in the span.

Source reference. Original lemma 08XA: sources/stacks-chow/tags.txt, source lines 11298–11298. Navigation pointer for \ref{descent-theorem-descent} used inside the counted span (line 349); the argument that consumes it is retained in the span.

Source reference. Original lemma 041Y: sources/stacks-chow/tags.txt, source lines 5309–5309. Navigation pointer for \ref{spaces-descent-lemma-descending-property-isomorphism} used inside the counted span (line 359); the argument that consumes it is retained in the span.

Source reference. Original lemma 03H6: sources/stacks-chow/tags.txt, source lines 4555–4555. Navigation pointer for \ref{spaces-properties-lemma-quasi-compact-affine-cover} used inside the counted span (line 333); the argument that consumes it is retained in the span.

Source reference. Original lemma 03M3: sources/stacks-chow/tags.txt, source lines 4735–4735. Navigation pointer for \ref{spaces-properties-proposition-quasi-coherent} used inside the counted span (line 338); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a quasi-compact monomorphism of algebraic spaces such that for every $T \to Y$ the map*

$$\mathcal{O}_T \to f_{T,*}\mathcal{O}_{X \times_Y T}$$

is injective. Then f is an isomorphism (and hence representable by schemes).

Proof. The question is étale local on Y, hence we may assume $Y = \mathrm{Spec}(A)$ is affine. Then X is quasi-compact and we may choose an affine scheme $U = \mathrm{Spec}(B)$ and a surjective étale morphism $U \to X$ (Properties of Spaces, Lemma 03H6). Note that $U \times_X U = \mathrm{Spec}(B \otimes_A B)$. Hence the category of quasi-coherent $\mathcal{O}_X$-modules is equivalent to the category $DD_{B/A}$ of descent data on modules for $A \to B$. See Properties of Spaces, Proposition 03M3, Descent, Definition 023G, and Descent, Subsection 08WW. On the other hand,

$$A \to B$$

is a universally injective ring map. Namely, given an A-module M we see that $A \oplus M \to B \otimes_A (A \oplus M)$ is injective by the assumption of the lemma. Hence $DD_{B/A}$ is equivalent to the category of A-modules by Descent, Theorem 08XA. Thus pullback along $f : X \to \mathrm{Spec}(A)$ determines an equivalence of categories of quasi-coherent modules. In particular f^* is exact on quasi-coherent modules and we see that f is flat (small detail omitted). Moreover, it is clear that f is surjective (for example because $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is surjective). Hence we see that $\{X \to \mathrm{Spec}(A)\}$ is an fpqc cover. Then $X \to \mathrm{Spec}(A)$ is a morphism which becomes an isomorphism after base change by $X \to \mathrm{Spec}(A)$. Hence it is an isomorphism by fpqc descent, see Descent on Spaces, Lemma 041Y. □